

Tutorial for Cyber-Physical Systems - Discrete Models

Exercise Sheet 12

Exercise 1*: Lecture Evaluation

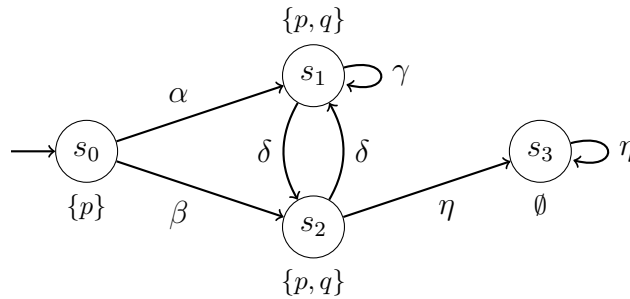
2 Bonus Points

Complete the lecture evaluation.

Exercise 2: Fairness

12 Points

Consider the following transition system:



and the following fairness assumptions:

F_1 Unconditional fairness for $\{\alpha\}$

F_2 Weak fairness for $\{\delta\}$

F_3 Strong fairness for $\{\gamma\}$

F_4 Weak fairness for $\{\delta\}$ and strong fairness for $\{\eta\}$

For each fairness assumption $F_1 - F_4$, answer the following questions. Justify your answer.

- (a) Is $s_0 \xrightarrow{\beta} s_2 \xrightarrow{\eta} s_3 (\xrightarrow{\eta} s_3)^\omega$ a fair execution?
- (b) Is $s_0 \xrightarrow{\alpha} s_1 \xrightarrow{\delta} s_2 (\xrightarrow{\delta} s_1 \xrightarrow{\delta} s_2)^\omega$ a fair execution?
- (c) Is $\{p\} \{p, q\}^\omega$ a fair trace?
- (d) Is $s_0 s_2 (s_1)^\omega$ a fair path?
- (e) Can the initial path fragment $s_0 s_2 s_1$ be extended to a fair path?
- (f) Is the fairness assumption realizable for the given transition system?

Exercise 3: Fairness

6 Points

For each of the following three systems (each consisting of one single process) give the weakest fairness assumption on action *enter* to ensure non-starvation. Non-starvation means that a process that has requested will eventually enter its critical section.

Give an informal explanation for your answers. Explain both why your chosen fairness assumption is *sufficient* to ensure non-starvation, and why it is the *weakest* possible assumption.

